



BRAC University
Department of Mathematics and Natural Sciences
MAT 216: Linear Algebra & Fourier Analysis
Mid Term Examination
Date: 04 November, 2023

Total Time : 1 hr 20 mins

FALL 2023

Total Mark: 40

Answer any 4 sets of questions out of 6 sets.
Please return the question with the answer script.

1. (a) Use Gauss-Jordan elimination (*equivalent to Reduced Row Echelon Form*) to solve the following system of linear equation. (6)

$$\begin{array}{rrrrr} -3x_2 & +2x_3 & & +x_4 & = -1 \\ -x_1 & +x_2 & -2x_3 & +7x_4 & = -7 \\ x_1 & & +2x_3 & -10x_4 & = 10 \end{array}$$

- (b) For what values of λ the following system of linear equation

$$\begin{array}{rrrr} x & +y & -z & = 1 \\ 2x & +3y & +\lambda z & = 3 \\ x & +\lambda y & +3z & = 2 \end{array}$$

has (i) a unique solution (ii) infinitely many solutions and (iii) no solution. (4)

2. (a) Calculate the inverse of the matrix $A = \begin{bmatrix} 3 & 0 & 2 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{bmatrix}$ (i.e. A^{-1}) by using $[A|I]$. (6)

- (b) Are the vectors $\vec{v}_1 = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ linearly independent? (4)

[Hints: Check the solution of the system $c_1\vec{v}_1 + c_2\vec{v}_2 = \vec{0}$]

3. **Vector Space:** A set V equipped with two binary operations addition and scalar multiplication is called a vector space over the field $\mathbb{F}$, if V satisfies the following 10 axioms,

- $\mathbf{u} + \mathbf{v} \in V$ for all $\mathbf{u}, \mathbf{v} \in V$
- $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
- $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
- There exists a $\mathbf{0} \in V$ s.t. $\mathbf{u} + \mathbf{0} = \mathbf{u}$ for all $\mathbf{u} \in V$
- There exists a $-\mathbf{u} \in V$ for all $\mathbf{u} \in V$, such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
- $k\mathbf{u} \in V$ for all $\mathbf{u} \in V$ and $k \in \mathbb{F}$
- $(ab)\mathbf{u} = a(b\mathbf{u})$ for all $a, b \in \mathbb{F}$ and $\mathbf{u} \in V$
- $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$ for all $a \in \mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in V$
- $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$ for all $a, b \in \mathbb{F}$ and $\mathbf{u} \in V$
- $1\mathbf{u} = \mathbf{u}$, where $1 \in \mathbb{F}$ and for all $\mathbf{u} \in V$.

- (a) Check whether the set of vector $V = \{\vec{x} \mid I\vec{x} = \vec{0}\}$ is a vector space over the field of real number $\mathbb{R}$ and under the standard operation defined for addition and multiplication, where, I is the $n \times n$ **identity matrix**. (5)

- (b) Is the set of vectors $W = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid y = -4x - z, z = 8x \right\}$ a subspace of the vector space $V = \mathbb{R}^3$? (5)

4. Consider the following matrix,

$$A = \begin{bmatrix} 1 & -3 & 0 & 4 \\ -1 & 4 & -2 & 5 \\ -2 & 6 & 0 & -8 \end{bmatrix}$$

- (a) Calculate the basis of the $Row(A)$. Also calculate the rank of A. (4+1)

- (b) Calculate the basis of the $Null(A)$. Also calculate the nullity of A. (4+1)

5. (a) Consider the linear transformation $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$, where,

$$T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{bmatrix} x - 2y \\ y - x + z \end{bmatrix}$$

Calculate the $\dim(Img(T))$ and $\text{rank}(T)$. (5+1)

- (b) Show that the vectors $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$, and $\vec{v}_3 = \begin{bmatrix} 3 \\ 0 \\ -4 \end{bmatrix}$ span $\mathbb{R}^3$. (4)

6. Find the eigenvalues and corresponding eigenvectors of the matrix, (10)

$$A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$$

Best wishes